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Module - 3

Web Development

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Chapters for Discussion

UNIT 1 — Introduction to HTML

UNIT 2 — CSS Fundamentals

UNIT 3 — Responsive CSS

UNIT 4 — Advanced CSS

UNIT 5 — JavaScript

UNIT 6 — JavaScript Interactivity

CHAPTER 1

Introduction to HTML

Learning Outcome

You will learn in this Chapter:

- Understand the basics of HTML
- Learn Various HTML tags
- Create and implement the structure of Web Pages

What is HTML?

- HTML stands for Hyper Text Markup Language
- HTML is used to create and design web pages and web applications.
- It describes the structure of a Web page
- HTML consists of a series of elements. Each elements are represented by HTML Tag.

HTML Tag Symbol: < TagName >

- Each HTML tag has a specific meaning.
- HTML page extension is .html (File Name: example.html)

Source : https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcT6935wo8bLZh5FeafJEffqWKDOpNpx6UE5bg&s

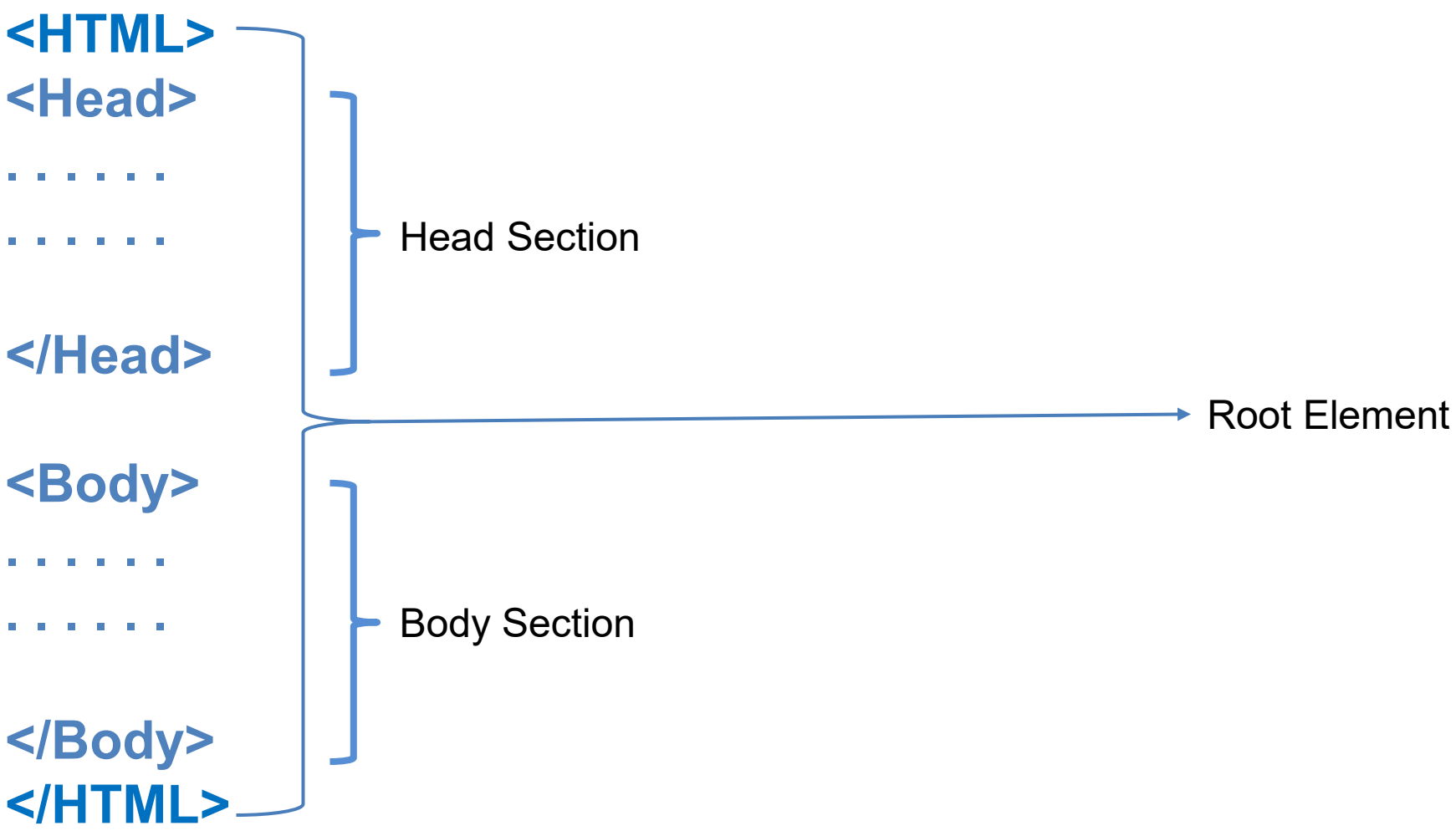
HTML Element

< TagName > Content < / TagName >

Opening Tag Closing Tag

- HTML is not case sensitive
- All HTML documents consist of nested HTML elements
- A few HTML elements are known as Empty Elements

HTML Element



HTML Comment

HTML Comment

Single-Line Comment

< ! - - Comment - - >

Points on Why Comments

- Non-executable lines of code
- Add readability
- Helps in debugging

Multi-Line Comment

< ! - - Comment line 1
Comment line 2
- - >

Points need to be taken care of

- Comments do not support nesting
- Can't have "- - " as content within a comment
- No spaces at the start of the comment string

HTML Basic Tags

Tag	Tag Name	Description
<TITLE> … </TITLE>	Title Tag	Used to add name to the web page which will be showcased on browser tab
<P> … </P>	Paragraph Tag	Used to add a paragraph on a web page
<H1> … </H1> <H2> … </H2> <H3> … </H3> <H4> … </H4> <H5> … </H5> <H6> … </H6>	Heading Tags	Used to create headings for the content of a webpage H1 – Top-level Heading [Largest font] H6 – Lowest-level Heading [Smallest font]
 	Break Tag	Used to break the line. The content will start adding from a new line.
<HR>	Horizontal Tag	Used to add a horizontal line to make a section / separation
<PRE>	Preformatted Tag	Used to add pre-formatted text content on a web page

HTML Formatting Tags

Tag	Tag Name	Description
<b>	Bold Tag	Used to add bold format on content
<u>	Underline Tag	Used to add underline to content
<i>	Italic Tag	Used to add italic format on content
	Emphasis Tag	Used to add emphasis to content
	Strong Tag	Used to add importance to content
<ins>	Insert Tag	Used to indicate that the content is been added on page
<sub>	Subscript Tag	Used to make subscript text (slightly below the normal line)
<sup>	Superscript Tag	Used to make superscript text (slightly above the normal line)
<mark>	Mark Tag	Used to highlight text with a yellow background
<del>	Delete Tag	Used to indicate that content has been deleted (typically struck through)

Lab Activity

Create a simple HTML page with headings, paragraphs, and line breaks for Portfolio Website

HTML List

HTML List
<li>

Ordered List
<ol>

1. One
2. Two
3. Three

Type Attribute: 1
a
i
A
I

Unordered List
<ul>

- One
- Two
- Three

Type Attribute: Disc
Circle
Square

HTML Nested List

```
<ol>
  <li>Coffee</li>
  <ul>        <li>Hot Coffee</li>
              <li>Cold Coffee</li>      </ul>
  <li>Tea</li>
  <ul>        <li>Black tea</li>
              <li>Green tea</li>        </ul>
  <li>Milk</li>
  <ul>        <li>Plain Milk</li>
              <li>Flavoured Milk</li>   </ul>
</ol>
```

1. Coffee
    - Hot Coffee
    - Cold Coffee
2. Tea
    - Black tea
    - Green tea
3. Milk
    - Plain Milk
    - Flavoured Milk

Lab Activity

Create ordered and unordered lists to represent your Technical skills & Achievements in Portfolio Website.

HTML Image

- HTML images provide visual content for web pages, enhancing user experiences and conveying information. They can be photographs, graphics, icons, or illustrations
- **<img>** tag, called the image tag, is used to add images to an HTML page
- It is an empty tag.

Syntax: < img src = “ path “ alt = “ text “ / >

Attribute Name	Value	Description
src	Path	Defines the path of the image (image URL)
alt	Alternative text	Defines the alternate text; if there is a broken link to the image path, the alternate text displays on the webpage
width	Numeric value in pixel	Define the height and width for the image
height		
border	Numeric value	Defines the border around the image

HTML Link

- The HTML **<a>** tag defines a hyperlink
- It is called an Anchor Tag
- It falls under the paired tag category as it consists of opening & closing tag

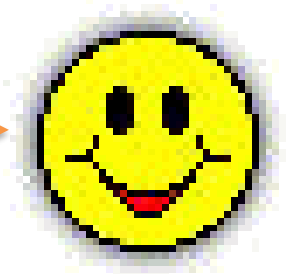
Syntax: < a href = “ URL “ > Click Here < /a >

Attribute Name	Value	Description
href	URL	Specifies the page/document to be linked
target	_blank, _self, targetframe	Specifies the location where linked document is opened

HTML Link (Conti..)

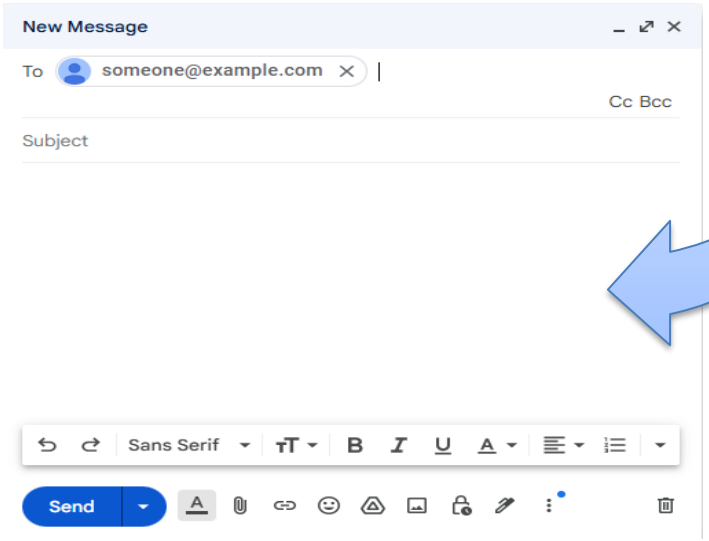
Image As Link

Click on Smiley



Email Link

Click on Send Email → Send Email



Lab Activity

Add hyperlinks and email links using the <a> tag for navigation in Portfolio Website.

HTML Table

- <table> tag is used to add data/content in tabular format on web page
- A table in HTML consists of table cells inside rows and columns.

Tag	Tag Name	Description
<TH>	Table Header	Specifies the header of table
<TR>	Table Row	Specifies the rows of table
<TD>	Table Data	Specifies the cell within a row of table

Attribute Name	Value	Description
border	Numeric Value	Used to put a border across all the cells

HTML Table (Conti..)

- HTML tables can have cells that span over multiple rows and/or columns
- These attributes can be used within <TH> or <TD> tags

Attribute Name	Value	Description
colspan	Numeric Value	Used to make a cell span over multiple columns
rowspan		Used to make a cell span over multiple rows

<table>
<tr><th colspan="2">NAME</th><th></th></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

<table>
<tr><td rowspan="3">APRIL</td><td></td><td></td></tr>
<tr><td></td><td></td></tr>
<tr><td></td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

<table>
<tr><th colspan="3">2022</th></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td colspan="2" rowspan="2">FIESTA</td><td></td></tr>
<tr><td></td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

Lab Activity

Design an HTML table using <table>, <tr>, and <td> for structured data like Educational Qualification in Portfolio Website

HTML Form

- An HTML form is a webpage section typically used for collecting data from users and then sending it to a server for further processing.
- The HTML **<form>** element defines a form to collect user information in HTML document

< label >	To add a text label to a field
< input >	Used to collect data from users in different types like textbox, radio button, password, email, etc.
< button >	Defines a clickable button
< textarea >	Defines a multi-line input field
< fieldset >	Used to group related data in a form
< select >	Used to create a drop-down list

HTML Form (Conti..)

<label>

- It is an HTML form element used to add a text label for an input field
- It is useful for screen-reader users

Syntax: < label > Label Name < / label >

Attribute Name	Value	Description
for	Element Id	Specifies the id of the form element the label should be bound to
form	Form Id	Specifies which form the label belongs to

HTML Form (Conti..)

<input>

- It specifies an input field where the user can enter data.
- It is an empty tag

Syntax: < input type = “ “ / >

Attribute Name	Value	Description
type	text	Used to add text box
	password	Used to add password box
	radio	Used to add radio buttons
	checkbox	Used to add check boxes

Attribute Name	Value	Description
type	date	Used to add calender
	file	Used to add file upload option
	submit	Used to add submit button
	reset	Used to add reset button

HTML Form (Conti..)

<textarea>

- This tag defines a multi-line text input control.
- It is often used in a form, to collect user inputs like comments or reviews.

Syntax: < textarea rows = “ “ cols = “ “ >
< / textarea >

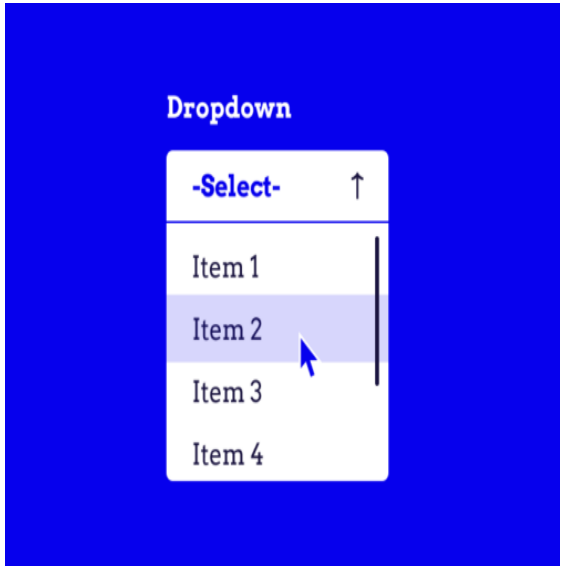
Attribute Name	Value	Description
name	text	Specifies a name for a text area
rows	Numeric Value	Specifies the visible number of lines in a text area
cols		Specifies the visible width of a text area

HTML Form (Conti..)

<select>

- This tag is used to create a drop-down list.
- The <option> tag inside the select tag defines the available options in the drop-down list.
- The <optgroup> tag inside select tag is used to group the options in one category

Syntax: < select >
< option > Item 1 < / option >
< option > Item 2 < / option >
< / select >



Tag Name	Attribute Name	Value	Description
<select>	size	Numeric value	Defines the number of visible options in a drop-down list
<option>	value	text	Specifies the value to be sent to a server

Source : https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTjewO0ZPyiyezLCS8KwBNw__aWdnwOYYquZRSM6_D0QQCa3TimNdxh6dnYSZZMmfN4Gh8&usqp=CAU

HTML Form (Conti..)

<button>

- This tag defines a clickable button

Syntax: < button > Name of Button < / button >

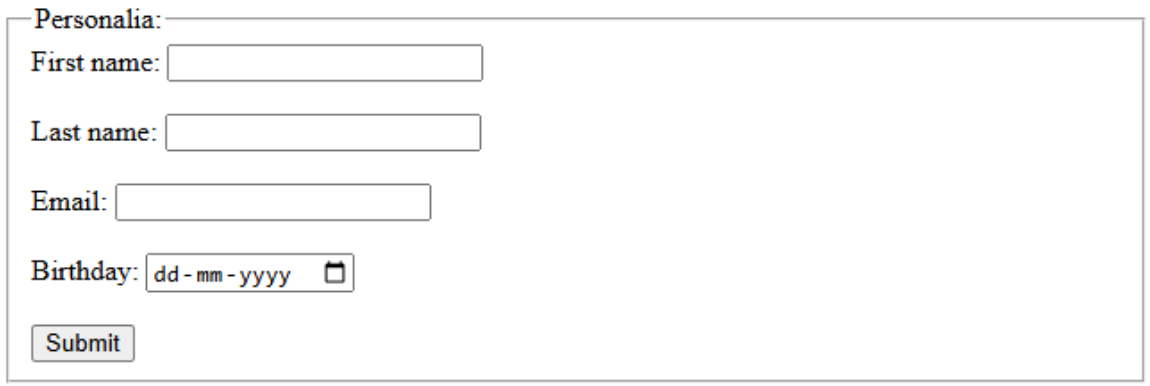
Attribute Name	Value	Description
name	text	Specifies a name for a button
type	Button, Submit, Reset	Specifies the type of button
value	text	Specifies an initial value for the button

Source : https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcR1GBmW47gvic6xMmeRqsVEIBiNlVq1E6jKlQ&s

HTML Form (Conti..)

<fieldset>

- It is used to group related elements in a form
- It draws a box around the related elements
- <legend > tag is used inside the fieldset tag to provide a name or title to the field



Syntax:

```
< fieldset >
        < legend >
                Name of field
        < / legend >
< / fieldset >
```

Source : https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcR1GBmW47gvic6xMmeRqsVEIBiNlVq1E6jKlQ&s

HTML Form (Conti..)

Form Attributes

Syntax: < form action = “ “ method = “ “ target = “ “ >

Attribute Name	Value	Description
action	URL	Defines the action to be performed when the form is submitted
	path	
method	get	specifies the HTTP method to be used when submitting the form data
	post	
target	_blank	specifies where to display the response that is received after submitting the form
	_self	
	framename	
autocomplete	on	Specifies whether a form should have autocomplete on or off
	off	

Lab Activity

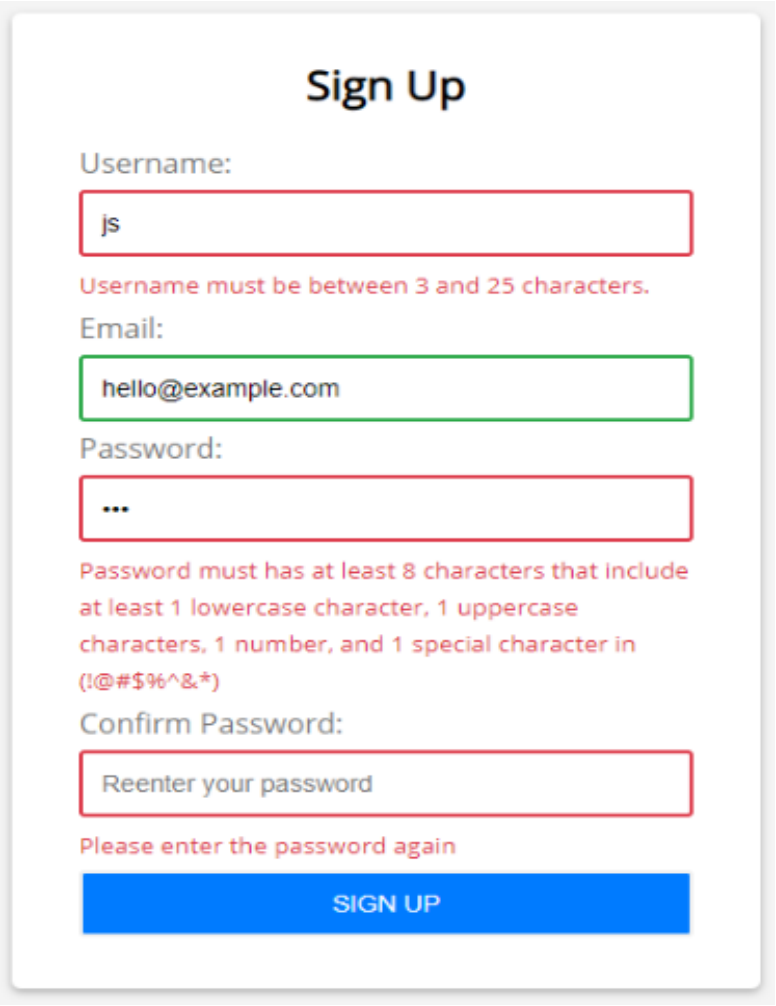
Create an interactive form using <input>, <button>, <select>, and <textarea> to capture feedback and questions from users of Portfolio Website.

HTML Form Validation

- HTML provides built-in mechanisms for client-side form validation, ensuring user input meets specific criteria before submission.
- This validation primarily relies on HTML5 attributes, which trigger browser-level checks and provide immediate feedback to the user.

There are different types of validation

- Required Field Validation
- Pattern Validation
- Range Validation
- Comparison Validation



Source : https://www.javascripttutorial.net/wp-content/uploads/2020/09/JavaScript-Form-Validation-Example.png

HTML Form Validation (Conti..)

Attribute Name	Value	Description
placeholder	text	Specifies a short hint that describes the expected value of a text area
required	required	Specifies that a text area is required/must be filled out
max	Numeric Value	Specifies the maximum value for an <input> element
min		Specifies a minimum value for an <input> element
minlength		Specifies the minimum number of characters required in an <input> element
maxlength		Specifies the maximum number of characters required in an <input> element

Lab Activity

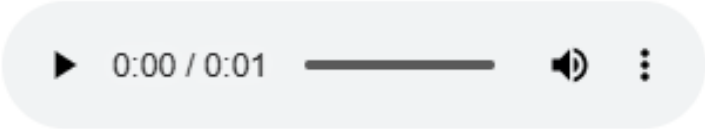
Validate a form using built-in attributes like required, pattern, and maxlength.

HTML Media Tag

Audio

- The HTML <audio> element is used to play an audio file on a web page
- The <source> element allows you to specify audio files

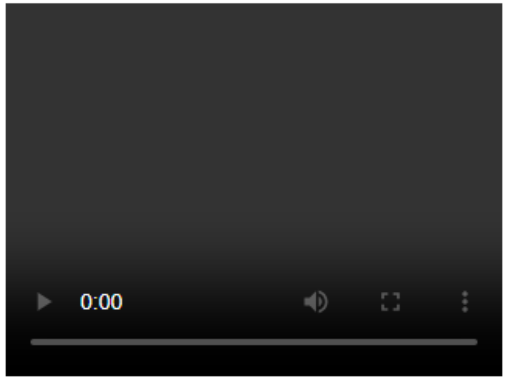
Syntax: < audio controls >
< source src = “ “ >
< / audio >



Video

- The HTML <video> element is used to show a video on a web page.
- The <source> element allows you to specify audio files

Syntax: < video controls >
< source src = “ “ >
< / video >



HTML Media Tag (Conti..)

Tag	Attribute Name	Value	Description
audio / video	controls	controls	Adds video controls, like play, pause, and volume
audio / video	autoplay	autoplay	Start playing automatically
audio / video	muted	muted	Keeps video or audio muted by default
video	height	Numeric Value	Used to set the height of video window
video	width	Numeric Value	Used to set the width of video window
source	src	Path	Used to add video / audio file
source	type	video/mp4	Used to provide type of file
		video/ogg	
		audio/mp3	
		Audio/ogg	

Lab Activity

Embed media files by adding images, audio, and video to showcase your talent, profile photo & video resume for Portfolio Website.

Div, Frame & Iframe Tag

Div Tag

- It is known as a division tag
- Used as a container for other HTML elements
- Used to group sections of a web page together

Syntax:

** . . . **

Frame Tag

- Frame tag is a child tag of Frameset tag
- Used in HTML4. It is not there in HTML5

Syntax:

<frameset>

<frame name=“ “ src=“ “ />

</frameset>

Iframe Tag

- It is known as Inline Frame
- Allows you to embed another document within the current HTML document

Syntax:

<iframe name=“ “ src=“ “ >

</iframe>

Semantic HTML

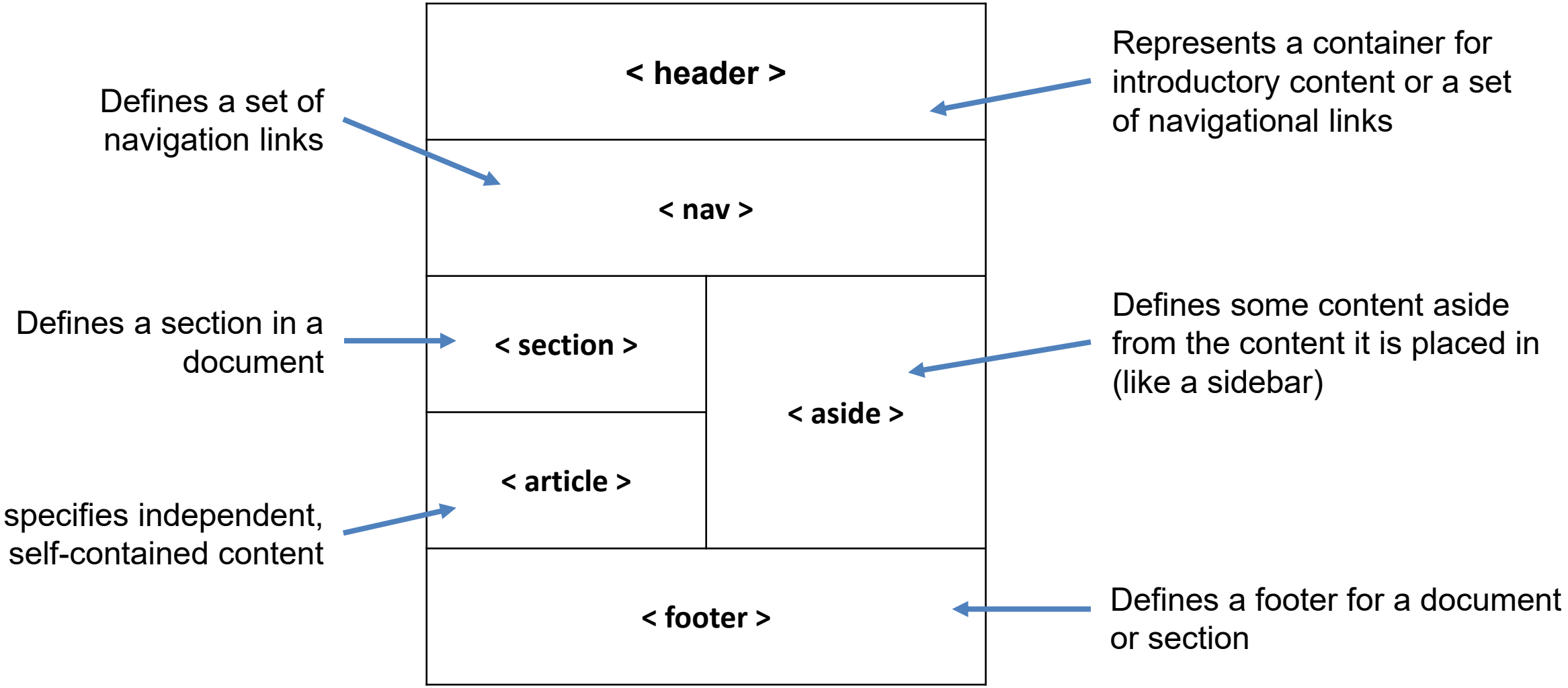
- Semantic HTML refers to the practice of using HTML elements to structure web content based on their inherent meaning and purpose, rather than solely on their visual presentation.
- This means choosing the most appropriate HTML tag for the content it encloses, providing semantic meaning that can be understood by browsers, search engines, and assistive technologies.

Why Semantic Tags?

- Improved Accessibility
- Enhanced SEO
- Better Code Readability & Maintenance
- Separation of Concerns

< header >	< nav >	< article >
< section >	< aside >	< footer >

Semantic HTML (Conti..)



Semantic HTML (Conti..)

< section >

- Chapters
- Introduction
- News items
- Contact information

< article >

- Forum posts
- Blog posts
- User comments
- Product cards
- Newspaper articles

< header >

- one or more heading elements (<h1> - <h6>)
- logo or icon
- authorship information

< footer >

- authorship information
- copyright information
- contact information
- sitemap
- related documents

< nav >

- Menu (navigation list)

< aside >

- Sidebars
- Pull quotes
- Advertisements

Lab Activity

Explore HTML5 semantic elements such as <header>, <footer>, <article>, <section>, and <aside> for Portfolio Website

HTML Component

HTML Component

Block Component

- Always starts on a new line
- Browsers automatically add some space (a margin) before and after the element.

- <p> <h1>-<h6>
- <ol> <header> <article>
- <ul> <section> <footer>
- <li> <pre> <aside>
- <nav> <table> <hr>
- <video>

Inline Component

- Does not start on a new line
- Only takes up as much width as necessary.

- < a > <sub> <sup>
-
 <img> < b >
- < u > < i > <input> <label>
- <select>
- <textarea>

HTML Accessibility Guidelines

Aim is to make web content usable by a wide range of individuals, including those with disabilities, by adhering to principles of perceivability, operability, understandability, and robustness.

- Semantic HTML
- Keyboard Accessibility
- Clear Language and Structure
- Transcripts and Captions
- Descriptive Links
- Accessible Forms
- Alternative Text for Images
- Language Declaration

HTML Meta Tag

- The <meta> tag in HTML is used to provide metadata about an HTML document.
- Metadata is data that describes other data, and in the context of a webpage, it provides information to browsers, search engines, and other web services about how to handle or display the content.

Character Set Declaration	Syntax: <meta charset="UTF-8">
Page Description	Syntax: <meta name="description" content="...">
Keywords	Syntax: <meta name="keywords" content="...">

HTML Meta Tag (Conti..)

Author Information

Syntax: <meta name="author" content="...">

Viewport Settings

Syntax: <meta name="viewport" content="width=device-width, initial-scale=1.0">

HTTP Equivalents

Syntax: <meta http-equiv="refresh" content="5">

HTML Entities

HTML entities are special sequences of characters used in HTML to represent characters that are either reserved in HTML syntax, not readily available on a standard keyboard, or are invisible characters.

Result	Description	Name
	Non - breaking Space	
<	Less Than	<
>	Greater Than	>
&	Ampersand	&
“	Double Quotation Mark	"
‘	Single Quotation Mark	'
©	Copyright	&copy

Lab Activity

Apply accessibility features using meta tags, HTML entities

HTML Debugging

- Debugging HTML primarily involves identifying and resolving issues within the structure and content of your web page.
- While HTML itself is not a programming language that executes logic in the same way as any programming language, errors in HTML can lead to unexpected rendering, layout problems, or broken functionality.

Browser Developer Tools

- Inspect Element/Elements Panel
- Console

Code Examination and Validation

- Manual Code Review
- HTML Validators

Code Formatting and Indentation

Cross-Browser Testing

Lab Activity

Debug HTML code by identifying and fixing common errors using browser tools.

Summary

You have learned the following topics in this Chapter:

- What is HTML?
- Basics of HTML
- HTML Elements
- Semantic HTML
- How to debug HTML

Reference

- https://developer.mozilla.org/en-US/docs/Learn_web_development/Core
- https://www.w3schools.com/html
- https://www.geeksforgeeks.org/html
- https://www.tutorialspoint.com/html

QUIZ

Quiz

1. **Which HTML tag is called the root element of an HTML document?**

a) <body>

b) <audio>

c) <html>

d) <ul>

Answer: C
<html>

Quiz

2. What tag is used to render an image on a webpage?

a) img

b) src

c) image

d) photo

Answer: A

<img>

Quiz

3. What is meant by an empty tag in HTML?

a) There is no such concept of Empty tag in HTML

b) An Empty tag does not require a closing tag

c) An Empty tag cannot have any content within it

d) None of Above

Answer: B

An Empty tag does not require a closing tag

Quiz

4. What is the select tag used for?

a) Creates a combo box

b) Select some attributes and changes their style

c) Change text font

d) Select a value

Answer: A
Creates a combo box

Quiz

5. HTML files are saved by default with the extension?

a) .html
b) .h
c) .http
d) .web

Answer: A
.html

Thank You